

ARYAN ZAVERI

Quant Developer | Trading Systems | Research Infrastructure | Execution

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SUMMARY

Quant Developer focused on end-to-end trading infrastructure: market data, cross-asset research, backtesting, replay, risk decisions, order managers and execution analytics. Python for research/backtesting; C++ for live socket, risk and order-runtime components; Redis hot state and PostgreSQL persistence.

SKILLS

Languages: Python, C++, SQL, Shell, JavaScript

Systems: Linux, AWS, Git, Docker, WebSockets, REST APIs, broker/exchange APIs, structured message contracts

Data: Redis streams/hashtables, PostgreSQL, SQLite, market data archives, candles, Greeks, snapshots, analytics outputs

Quant: feature engineering, cross-asset research, backtesting, walk-forward validation, replay, risk managers, order managers, execution journals

Markets: equities, futures, options, crypto; Indian/global sectors, oil, FX, macro, volatility and breadth/regime context

WORK EXPERIENCE

GreyOak Capital - Quantitative Developer | Mar 2026 - Present

- Build quantitative research systems, market-intelligence infrastructure, market-data services and analyst-facing context-intelligence tooling.
- Work on Horizon, Compass, Context-Intelligence and Ask-GreyOak style systems connecting market context, internal data flows, dashboards, WebSocket/data contracts and research interfaces.
- Own research-to-runtime handoffs across historical archives, live state, signal outputs, runtime logs, validation reports and production-readiness workflows.

KIFS Trade Capital - Quantitative / Algorithmic Trading Developer | 2024 - Mar 2026

- Built algorithmic trading tools, broker/API workflows, intraday market-data processing, conversion/reversion research utilities and execution analytics.
- Developed intraday/options analytics around tick/depth context, price action, option-chain features, IV/OI/Greeks-style inputs, signal filters and post-market validation.
- Worked on order lifecycle mechanics, execution constraints, latency-aware design and reusable net-PnL evaluation modules for trade-quality analysis.

Greeksoft Technologies - Algorithmic Trading Intern

- Worked around broker-facing algorithmic trading platform workflows: broker connectivity, instrument mapping, market data, order placement, order status and execution flow.
- Gained practical exposure to how broker APIs, authentication, market-data subscriptions and order-state transitions fit inside an algo-trading platform.

PYTHON RESEARCH SYSTEMS

- Built Python research/control-plane systems for feature engineering, signal generation, backtesting, walk-forward validation, replay, portfolio experiments and reports.
- Used pandas/numpy/sklearn-style pipelines across returns, volatility, sector breadth, futures basis, IV/OI/Greeks, macro context, BTC-relative and cross-asset features.
- Built options/equity/crypto research flows covering model training, threshold selection, fee/slippage-aware simulation, risk overlays, daily/monthly summaries and dashboards.
- Integrated Python with Redis/PostgreSQL for market archives, live feature tables, state writers, bridge services, monitor APIs, reports and operational jobs.

C++ TRADING RUNTIME

- Built C++ live components for market/private sockets, order execution, position tracking, risk management, signal bridges, hot-state daemons and replay/parity tooling.
- Used Boost.Asio/Beast/OpenSSL, nlohmann::json, CMake, threads, atomics, mutexes, hiredis and libpq-style integrations across live/runtime paths.
- Implemented order preflight and smart-execution flows: account/instrument probes, Redis quote checks, quote freshness, slippage gates, duplicate-intent checks and exchange journals.
- Designed low-latency state handling with Redis streams/hashtables, in-memory working-order state, heartbeats and durable PostgreSQL writes outside the execution path.

SELECTED PROJECT - VAJRA

- Built a personal multi-asset research/trading platform connecting Indian sectors, global indices, crude oil, FX, macro context, options/futures structure and crypto risk appetite.
- Implemented the research-to-execution loop: market data archive, feature engineering, signal ranking, risk decision, order intent, execution journal, replay/PnL analytics and model refinement.
- Built independent risk authority and adaptive execution around global capital, segment risk, quote freshness, spread/slippage gates, price adjustment, laddering and hard exits.

EDUCATION

B.Tech in Information Technology, Mukesh Patel School of Technology Management and Engineering (MPSTME), NMIMS.